

Simmon Yang

Toronto, ON

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SUMMARY

Detail-oriented Electrical Engineering Technology student with hands-on experience in electrical systems, low-voltage infrastructure, renewable energy hardware, and software development. Strong background in field installation, troubleshooting, and system integration, complemented by programming and automation experience. Seeking technical roles where electrical, systems, and problem-solving skills intersect.

TECHNICAL SKILLS

- **Electrical & Hardware:** DC/AC Circuits, Digital Circuits, Residential & Commercial Wiring, Low-Voltage Systems, Microcontrollers (Arduino, Raspberry Pi), PCB Design (KiCad), Security Systems, Network Cabling (Cat5e/Cat6/Cat6A), Soldering, Testing & Troubleshooting
- **Software & Tools:** Python, C, C++, Java, PHP, WordPress, MySQL, AutoCAD Electrical, Multisim, PLC Programming
- **Standards & Safety:** Electrical Code Comprehension, TIA/EIA Standards, WHMIS, Working at Heights, Lockout/Tagout

EDUCATION

Electrical Engineering Technology Diploma

Sep 2022 – Present

Centennial College

Scarborough, ON

- Relevant coursework: Electrical Theory, Drawing Interpretation & Installation, Monitoring Systems, Networking, PLC Programming, Power Generation, Microcontrollers, Signals & Communication
- GPA: 4.1

WORK EXPERIENCE

Electrician Assistant

Sep 2024 – Present

Independent / On-call

Greater Toronto Area, ON

- Assisted licensed electricians with installation, maintenance, and repair of residential and commercial electrical systems.
- Pulled, cut, and terminated wiring and conduit for lighting, receptacles, and power distribution.
- Installed switches, outlets, lighting fixtures, and breakers in compliance with electrical codes.
- Supported panel upgrades, rough-ins, grounding, and device installations.
- Performed continuity, voltage, and insulation resistance testing using multimeters.

Low Voltage Technician

Jul 2025 – Sep 2025

OneNid Inc

Greater Toronto Area, ON

- Installed and terminated Cat5e/Cat6/Cat6A cabling for IP cameras, access points, and intercom systems.
- Mounted and calibrated IP and analog security cameras.
- Installed and configured DVR/NVR systems with network and remote access setup.
- Integrated access control systems including card readers, maglocks, and motion sensors.
- Ensured cabling and terminations met TIA/EIA and building code standards.

HVAC Assistant

May 2024 – Dec 2024

HVAC Contractor

Greater Toronto Area, ON

- Assisted with installation, maintenance, and repair of HVAC systems.
- Supported ductwork fabrication, copper piping, brazing, and insulation.
- Helped install and wire thermostats, compressors, condensers, and air handling units.
- Performed system checks, filter replacements, and refrigerant pressure monitoring.

Web Developer (Contract)

Mar 2024 – Apr 2024

Riipen

Remote

- Designed and developed a dynamic website using PHP and WordPress.
- Implemented backend logic for authentication and MySQL database integration.
- Built responsive layouts and deployed applications to live hosting environments.

Residential Renovation Assistant

May 2023 – Aug 2023
Greater Toronto Area, ON

Construction Inc

- Installed residential electrical wiring including switches, outlets, and lighting fixtures.
- Assisted with drywall installation, painting, and flooring demolition.

Operations & Systems Assistant

Jun 2022 – Aug 2022

Microgreen Solar Corp

Markham, ON

- Coordinated inventory management and shipment logistics.
- Developed a WordPress/PHP-based inventory management system with MySQL integration.

Electrical Technician Assistant

Jul 2021 – Aug 2021

Microgreen Solar Corp

Markham, ON

- Repaired and tested DC-AC inverter power packs for renewable energy systems.
- Performed component-level troubleshooting and soldering.
- Conducted load testing and quality assurance verification.

PROJECTS

Automated Circadian Lighting System (Retrofit)

Sep 2025 – Dec 2025

Lighting Control / Energy Efficiency

- Designed a retrofit automated LED lighting system to improve human well-being and energy efficiency by aligning indoor lighting with natural circadian rhythms.
- Developed a control strategy to dynamically adjust light intensity and correlated color temperature throughout the day, providing cooler, brighter light during daytime and warmer, lower-intensity light in the evening.
- Engineered the system to integrate with existing LED pot lights without replacing fixtures, minimizing installation cost and infrastructure changes.
- Analyzed power consumption and lighting performance to demonstrate improved visual comfort, circadian support, and reduced energy usage.

Adaptive LED Headlight Simulation

Sep 2025 – Dec 2025

PIC16F747 / Embedded Systems

- Designed a microcontroller-based adaptive automotive headlight simulation using a PIC16F747 and an HC-SR04 ultrasonic distance sensor.
- Implemented distance-based LED control logic that dynamically adjusted illumination patterns to simulate glare reduction for oncoming vehicles.
- Processed ultrasonic echo timing using internal timers and mapped measured distance to discrete LED activation zones.
- Programmed the system in C using MPLAB X and the XC8 compiler with polling-based sensor control and conditional output logic.
- Investigated PWM-based dimming strategies and identified timing and performance constraints causing LED flicker.
- Validated embedded control concepts including real-time sensing, hardware–software integration, and system-level trade-offs.

8x8x8 RGB LED Cube

Dec 2021 – Jun 2023

Arduino / PCB Design

- Designed and built a 3D RGB LED cube controlled by an Arduino microcontroller.
- Designed custom PCBs and 3D-printed enclosures for final implementation.

Pathfinding Visualization

Dec 2020 – Feb 2021

Python (Pygame)

- Developed an interactive visualization of A*, BFS, DFS, and Dijkstra algorithms.
- Implemented modular, object-oriented code for scalability.

CERTIFICATIONS

- WHMIS Certification (2024)
- Working at Heights Certification (2024)
- G3 Gas Technician (2025)